

MAM · Rectilinear Motion — Patterns and Solution Strategies

Based on: the marking keys + exam reports for 13 rectilinear motion questions from 2016–2025 (the 2018, 2024 and 2025 reports explicitly list rectilinear motion as a weak area; the 2021 report reminds students to distinguish distance from displacement; the 2023 report names signed area and the dt notation). Original questions are in [topics/03_Rectilinear_Motion.pdf](#). Review status: has been through one round of independent review and revised in line with the feedback (see [Review_log.md](#)).

Concept quick reference (the reports say these are the concepts most easily confused)

Quantity	Definition	Watch out for
Displacement $x(t)$	Position relative to the origin, positive or negative	'Returning to the starting point' is $x(t) = x(0)$, not $v = 0$
Velocity $v = x'(t)$	Positive or negative (direction)	'at rest' is $v = 0$; 'changes direction' is v changing sign
Speed $= v $	Non-negative	For maximum speed, compare $ v $ at the local extrema and at the endpoints
Acceleration $a = v'(t)$	Positive or negative	Where the velocity takes a local extremum inside the interval, $a = 0$ is guaranteed; the converse is not true (e.g. $v = t^3$ at $t = 0$; on a constant-velocity section a is always 0); endpoints are compared separately
Change in displacement $= \int_a^b v \, dt$	Signed area	'What does the signed area represent?' = change in displacement from $t=a$ to $t=b$
Distance travelled $= \int_a^b v \, dt$	Non-negative	Must be split into sections at $v = 0$

Speeding up or slowing down: v and a with the **same sign** is speeding up; with **opposite signs** is slowing down. The exact words in the 2024F-Q2 key: saying only that acceleration is negative is not a valid reason.

Note: the key for 2018A-Q11(c) treats "heading south and decelerating at 0.5" as $a = -0.5$ (by the definition above, v and a then have the same sign, so it is actually speeding up). With wording like this, state your interpretation clearly and specify the positive direction.

Time wording: 'when it first appears' and 'initially' mean $t = 0$ (2024 report: many students substituted $t = 1$ by mistake); 'during the fifth second' means $4 \leq t \leq 5$.

Pattern 1: Given $x(t)$, find v , a and what they mean

Approach: differentiate to get v ; differentiate again to get a ; substitute the time asked for; **write the units**; in your explanation state the direction clearly (what the positive direction is — for example north, up, to the right (in 2024F-Q2, x is the distance from the left edge of the screen; the key writes "to the right")).

Mark-scoring points: ✓ expression for $v(t)$ ✓ substituted value (with units) ✓ expression for $a(t)$ ✓ conclusion with reason (same sign/opposite signs).

Past questions: 2016F-Q4, 2016A-Q19(a), 2022A-Q10(b), 2024F-Q2(a)(b).

Pattern 2: Given $a(t)$ or $v(t)$, work backwards using the initial conditions

How to recognise it: 'initial velocity is 5 cm/s', ' $x(0) = 0$ ', 'After 2 seconds displacement is -50 cm', and you have to find the constants p , k .

Approach

1. Integrate $a \rightarrow v$ (+ c) $\rightarrow$ use $v(0)$ to determine c .
2. Integrate again, $v \rightarrow x$ (+ c_2) $\rightarrow$ use the known position to determine c_2 .
3. When the equation contains an unknown parameter p : write each of the two position conditions as an equation, then solve simultaneously for p and the constants (2017A-Q20).

4. For a trigonometric v (e.g. $v = 2 \sin(t/3 + \pi/6)$), when integrating divide by the inner coefficient $1/3$, i.e. multiply by 3, giving $-6\cos(t/3 + \pi/6) + C$; set your calculator to radians.

Mark-scoring points (2018A-Q11(a)): ✓ integrate to obtain the cos form ✓ write the constant and form an equation using $x(0) = 0$ ✓ find C and write down $x(t)$.

Past questions: 2017A-Q20(b), 2018A-Q11, 2019A-Q9(c), 2021A-Q17(a), 2022A-Q11(c), 2023A-Q8(e) (equate coefficients to find A, B).

Pattern 3: When it is at rest / changes direction / returns to the starting point

Approach

- At rest: solve $v(t) = 0$. Changes direction: v must **change sign** (check whether v really changes sign either side of that point; if v merely touches 0 and returns to the same sign, e.g. $v = (t - 2)^2$, the object has not changed direction).
- Returns to the starting point: solve $x(t) = x(0)$; passes through the origin: solve $x(t) = 0$ (these are the same only when $x(0) = 0$). Take the 'first' positive root the question asks for.
- When you only have a sign table (2025F-Q6): x changes sign $\rightarrow$ passes through the origin; v changes sign $\rightarrow$ stops and turns around; v and a with opposite signs $\rightarrow$ slowing down. In your explanation, state that the basis is 'the sign of v changes'.

Past questions: 2016A-Q19(a), 2017A-Q20(c), 2021A-Q17(b), 2022A-Q10(a)(c), 2023A-Q8(a), 2025F-Q6.

Pattern 4: Total distance travelled vs displacement ★ guaranteed to appear

How to recognise it: 'total distance travelled', 'How far has ... travelled' (these ask for distance) vs 'Evaluate $\int v \, dt$ and explain' (this asks for the change in displacement).

Standard procedure (total distance travelled)

1. Solve $v = 0$ to find all the times $t_1, t_2, \dots$ within the interval at which it stops.
2. Write the expression for the distance travelled; either form is acceptable:
 - $|x(t_1) - x(0)| + |x(t_2) - x(t_1)| + \dots$ (the form used in the key for 2024F-Q2(d));
 - $\int |v(t)| \, dt$, or integrate in sections: **put a negative sign in front of the section where $v < 0$** (or put absolute value signs around each section). In 2023A-Q8(c), $v = -2t \sin t$ is negative on $(\pi/3, \pi)$ and positive on $(\pi, 4\pi/3)$, so it is written as $-\int_{\pi/3}^{\pi} v \, dt + \int_{\pi}^{4\pi/3} v \, dt$. The key awards 1 mark separately for the signs, and 1 mark for **including dt on every integral**.
3. Calculate and include units.

Explaining $\int_a^b v \, dt$: 'the change in displacement (position) of ... from $t = a$ s to $t = b$ s' — you must also state the start and end times (in 2024F-Q2(c), 1 mark is awarded specifically for the start and end times).

Mark-scoring points: ✓ find the stopping times ✓ correct sectioned expression ✓ numerical value.

Common errors: treating $x(15) - x(0)$ directly as the distance travelled; missing the turning points in the middle when splitting into sections.

Past questions: 2016A-Q19(b)(d), 2017A-Q20(d), 2018A-Q11(c), 2021A-Q14, 2021A-Q17(c), 2023A-Q8(b)(c), 2024F-Q2(c)(d).

Pattern 5: Maximum velocity / zero acceleration

Approach: set $a(t) = v'(t) = 0$ and solve for $t \rightarrow$ substitute into $v \rightarrow$ **verify it is a maximum:** confirm using $v'' < 0$ (or a sign test); this step carries marks when the question says using calculus (2022A-Q11(b) key: confirms that $v'' < 0$); if the question asks for the maximum on the interval, also compare with the endpoints. If it asks for the maximum **speed**, also compare $|v|$ at the endpoints (2016A-Q19(c) key: examines velocity at endpoints).

Past questions: 2016A-Q19(c), 2022A-Q11(b), 2023A-Q8(d), 2021A-Q14, 2019A-Q12(c) (the first 2 minutes

are at constant velocity, a is always 0, so it is not an extremum).

Pattern 6: Interpreting context and graphs

How to recognise it: you are given a graph of $a(t)$ or $v(t)$ and asked to 'explain what is happening to the velocity in $0 < t < 8$ '.

How to write it: 'Since $a(t) > 0$ for $0 < t < 8$, the (upward) velocity is increasing; since $a(t) < 0$ for $8 < t < 16$, the velocity is decreasing (but still positive, still moving up).' Marking requirements: ✓ refer to the graph or sign of a ✓ conclusion for the first part ✓ conclusion for the second part.

Sketching (2025F-Q6(d)): the graph of $x(t)$ must satisfy all three sets of signs — x , v and a — at the same time: it starts at the origin with positive initial slope and crosses the t -axis only between $3 < t < 4$; the sign of the slope corresponds to v , and the concavity corresponds to a .

Past questions: 2019A-Q9(a)(b), 2025F-Q6(d), 2022A-Q10(d) (amplitude envelope $3e^{-t} = 0.01$).